

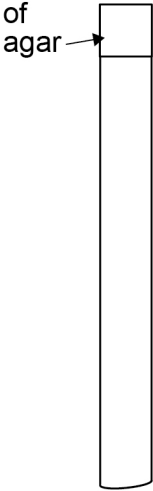
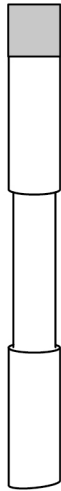
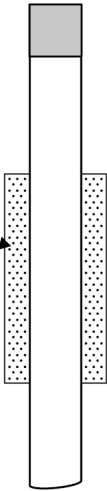
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Under the correct conditions, new roots grow from the cut end of a plant stem. A scientist investigated the effect of substance **X** on the growth of new roots.

She used a ringing experiment to investigate the movement of substance **X** in stems taken from lemon plants. She cut out a length of stem from each plant. She then put a small block of agar on the top of each length of stem. Some agar blocks contained substance **X**.

Figure 4 shows how she treated each length of stem.

Figure 4

D 	E 	F 	G 
No substance X in agar, middle section of stem intact	Substance X in agar, middle section of stem intact	Substance X in agar, middle section of stem ringed	Substance X in agar, middle section of stem intact and cooled to 4 °C

She grew the lengths of stem in the same environmental conditions for 6 weeks, and then found the number of roots per length of stem. Roots grew at the other end of the stem from where the agar blocks were placed.

Table 3 shows the scientist's results.

Table 3

Treatment	Mean number of roots per length of stem
D	5
E	11
F	4
G	3

Do not write
outside the
box



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Treatment **D** is a control. Explain how the measurement obtained from this control is used by the scientist.

[2 marks]

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Using **Figure 4** and **Table 3**, what can you conclude from treatments **D** and **E** about root growth?

[3 marks]

Turn over ►

The mass flow hypothesis is used to explain the movement of substances through phloem.

Evaluate whether the information from this investigation supports this hypothesis.
Do **not** consider statistical analysis in the answer.

[4 marks]

[illegible]

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